

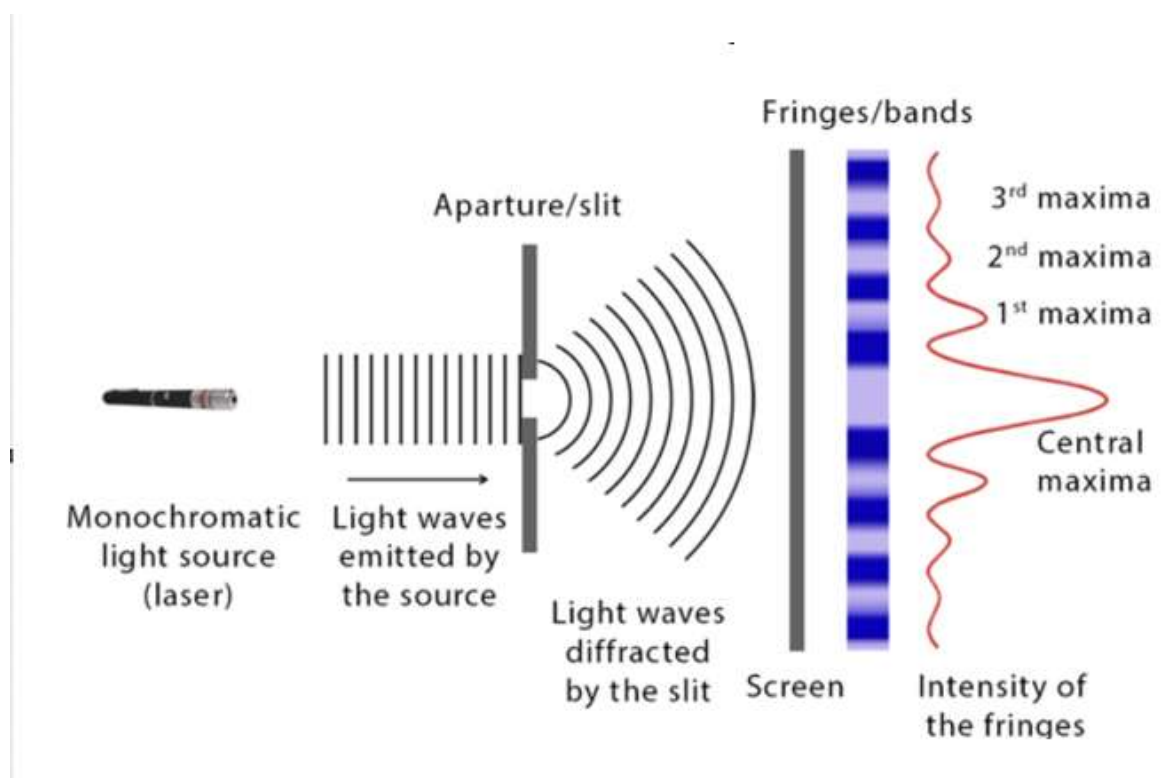
22PH202 ELECTROMAGNETISM AND MODERN PHYSICS

UNIT III LM-3.5

DIFFRACTION - INTERFERENCE –POLARISATION

1 DIFFRACTION

Diffraction is the bending or spreading of light waves around an obstacle. The obstacle can be an aperture or slit whose size is approximately the same as the wavelength of light. The amount of bending depends on the relative size of the opening compared to the wavelength of light. The bending will be almost unnoticeable if the opening is much larger than the wavelength. Due to diffraction, the direction of waves changes as they pass around an obstacle in their path.



Diffraction

According to Huygens' principle, the aperture or slit diffracting the waves becomes the secondary source of waves. The diffracted waves fall a screen and form a pattern known as a diffraction pattern. It consists of alternating dark and bright fringes spread across the screen. A bright central fringe is observed at the center, surrounded by several maxima and minima. Diffraction experiments have shown that light has wavelike properties.

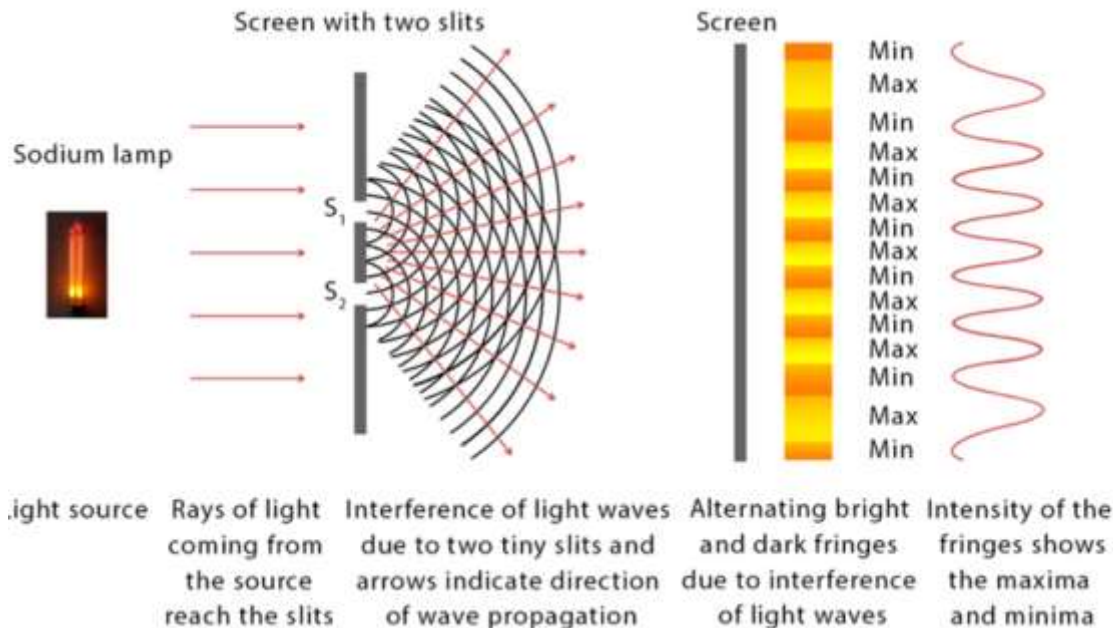
1.1 SINGLE-SLIT DIFFRACTION

Single-slit diffraction is the most straightforward experimental setup where diffraction effects can be observed. When light passes through a slit whose width is on the order of the wavelength of light, a distinct diffraction pattern is observed on a screen that is kept at a certain distance from the slit. The intensity is a function of the angle through which the rays bend.

All the waves passing through the slit interfere to produce a diffraction pattern consisting of bright and dark fringes. The bright fringes are due to constructive interference, and the dark areas are due to destructive interference. A number represents the order of the bright and dark fringes. The intensities of the fringes consist of a central maximum surrounded by maxima and minima on its either side. The central maximum is brighter than the other maxima. The maxima rapidly decrease as one moves further from the center.

1.2 DOUBLE-SLIT EXPERIMENT

The double-slit experiment is a test that demonstrates light can fundamentally display both wave and particle features, also known as wave-particle (photon) duality. It consists of two closely-spaced slits in front of a light source such that a beam of light passing through them is split and projected on a screen placed far from the slits. The split beam forms two coherent sources of light. A pattern of bright and dark fringes appears on the screen due to the interference of light waves coming from these two sources. Interference is the phenomenon in which when waves are superimposed, they may reinforce or cancel each other. It is most pronounced when the wavelength of the radiation is comparable to the linear dimensions of the slits.



Double-Slit Experiment

2 EXAMPLES AND APPLICATIONS OF DIFFRACTION

1. An example of diffraction is seen in the atmosphere. Diffracted light bends around atmospheric particles like tiny water droplets that are present in clouds producing light and dark fringes or colored bands. The diffraction of sunlight by clouds is a common phenomenon referred to as a silver lining.
2. Another example of diffraction is while observing the back of a compact disc (CD). Recorded data on CD is stored in tiny pits of different lengths, which carry the information. These pits have the same width and are equally spaced in a row, forming a diffraction grating on the CD mirror surface.
3. Hologram projection is also an example of diffraction. When two propagating waves cross each other's path, they create a standing wave. This standing wave, which remains still, can be photographed. By using two different waves of light, a standing wave is created and photographed, resulting in a hologram.
4. The appearance of the sun during sunrise and sunset is due to diffraction. The sun appears red because light gets diffracted in the presence of dust particles in the atmosphere. Also, when sunlight passes through trees, they are diffracted because trees are obstacles in the path of sunlight.
5. Other examples of diffraction include lights through a shadow of an object and the bending of light at the corners of the door.

3. INTERFERENCE

The two waves could add together either constructively or destructively. In constructive interference, the amplitude of the resultant wave is greater than that of either of the individual waves, whereas in destructive interference, the resultant amplitude is less than that of either individual wave. Light waves also interfere with each other. Fundamentally, all interference associated with light waves arises when the electromagnetic fields that constitute the individual waves combine. Interference effects in light waves aren't easy to observe because of the short wavelengths involved (about 4×10^{-7} m to about 7×10^{-7} m).

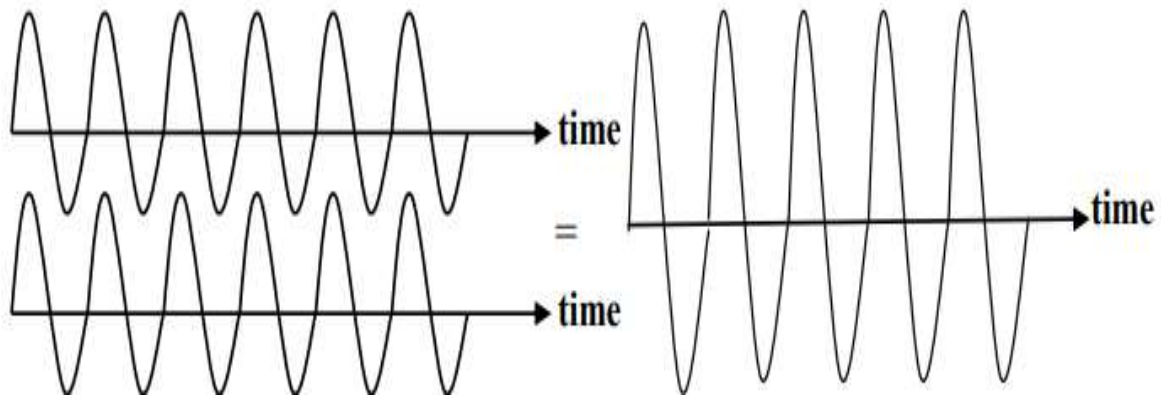


Interference

3.1 CONSTRUCTIVE INTERFERENCE

1. The sources must be coherent, which means the waves they emit must maintain a constant phase with respect to each other

2. The waves must have identical wavelengths. At certain points, the two waves may be in phase. The amplitude of the resultant wave will then be equal to the sum of the amplitudes of the two waves, as shown figure

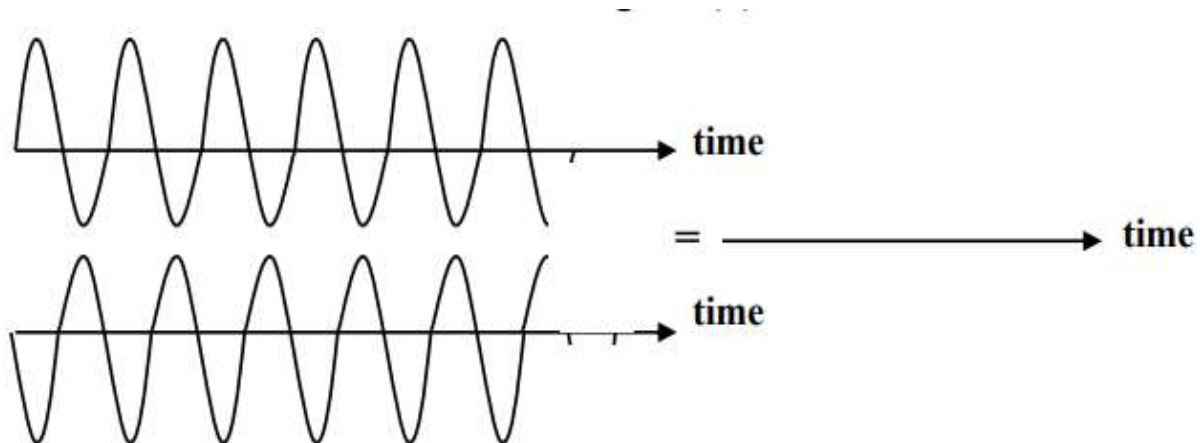


constructive interference.

Therefore, the interference produced at these points is known as constructive interference. A stationary bright band of light is observed at points constructive interference.

3.2 DESTRUCTIVE INTERFERENCE.

At certain other points, the two waves may be in opposite phase. The amplitude of the resultant wave will then be equal to the sum of the amplitudes of the two waves, as shown figure (2).



Destructive interference

Therefore, the interference produced at these points is known as destructive interference. A stationary dark band of light is observed at points destructive interference.

$$\Delta = m\lambda \quad \text{constructive interference}$$

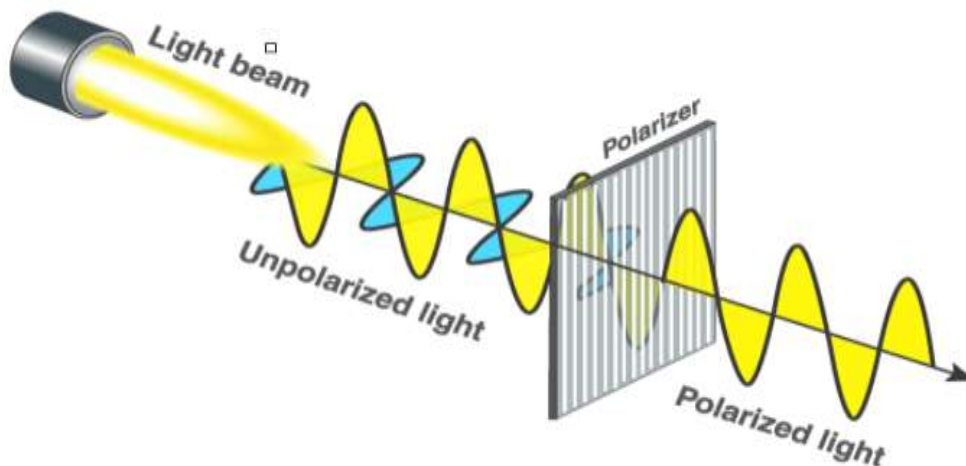
$$\Delta = (2m + 1)\frac{\lambda}{2} \quad \text{destructive interference}$$

Where $m=0, 1, 2, 3, \dots$ and Δ optical path difference.

The phenomenon of redistribution of light energy due to the superposition of light waves from two or more coherent sources is known as interference.

4. POLARIZATION

Light is the interaction of electric and magnetic fields travelling through space. The electric and magnetic vibrations of a light wave occur perpendicularly to each other. The electric field moves in one direction and the magnetic field in another 'perpendicular to each other. So, we have one plane occupied by an electric field, another plane of the magnetic field perpendicular to it, and the direction of travel is perpendicular to both. These electric and magnetic vibrations can occur in numerous planes. A light wave that is vibrating in more than one plane is known as unpolarized light. The light emitted by the sun, by a lamp or a tube light are all unpolarised light sources. As you can see in the image below, the direction of propagation is constant, but the planes on which the amplitude occurs are changing.



Polarization of Light

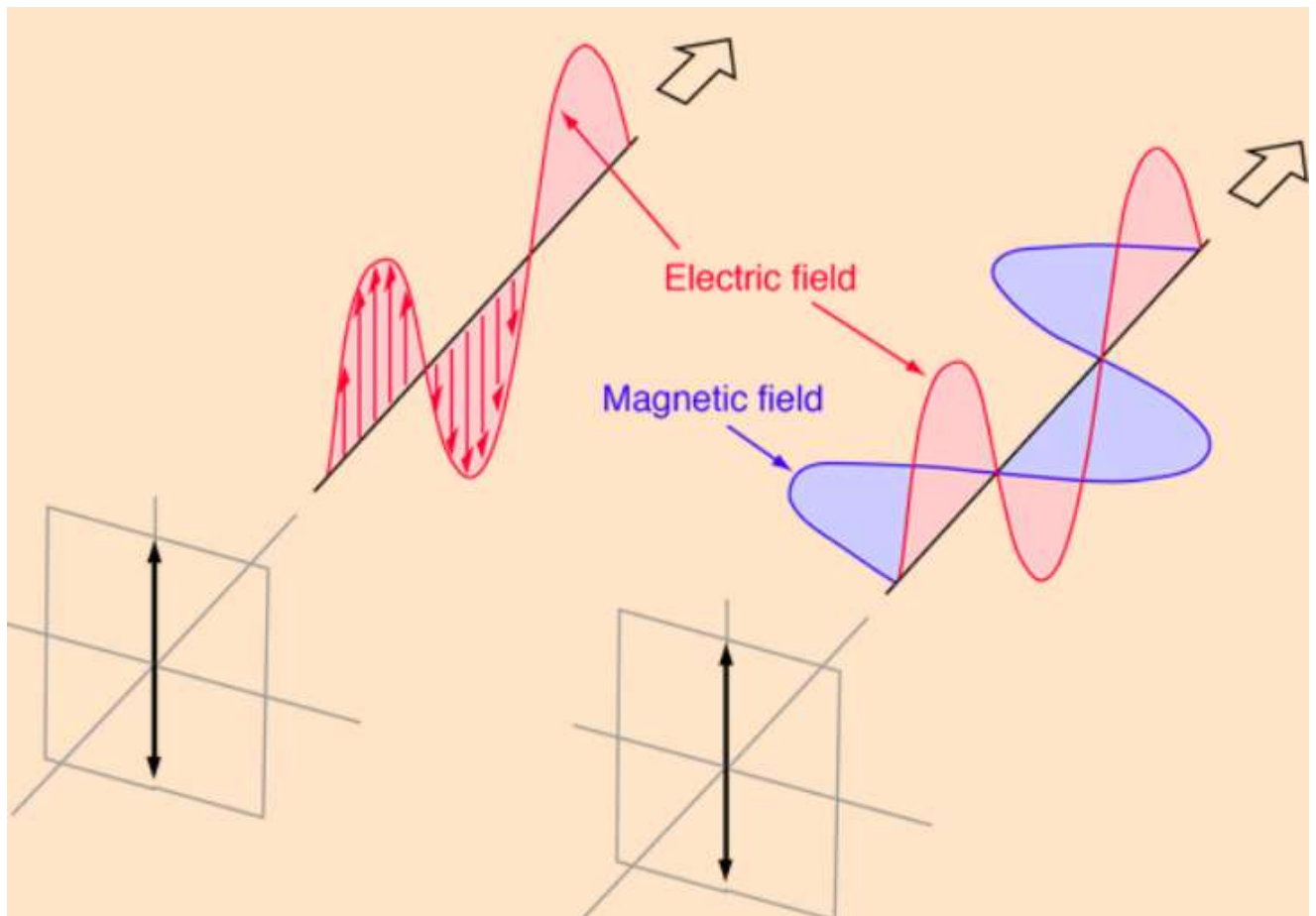
4.1 TYPES OF POLARIZATION

Following are the three types of polarization depending on how the electric field is oriented:

- Linear polarization
- Circular polarization
- Elliptical polarization

4.1.1 LINEAR POLARIZATION

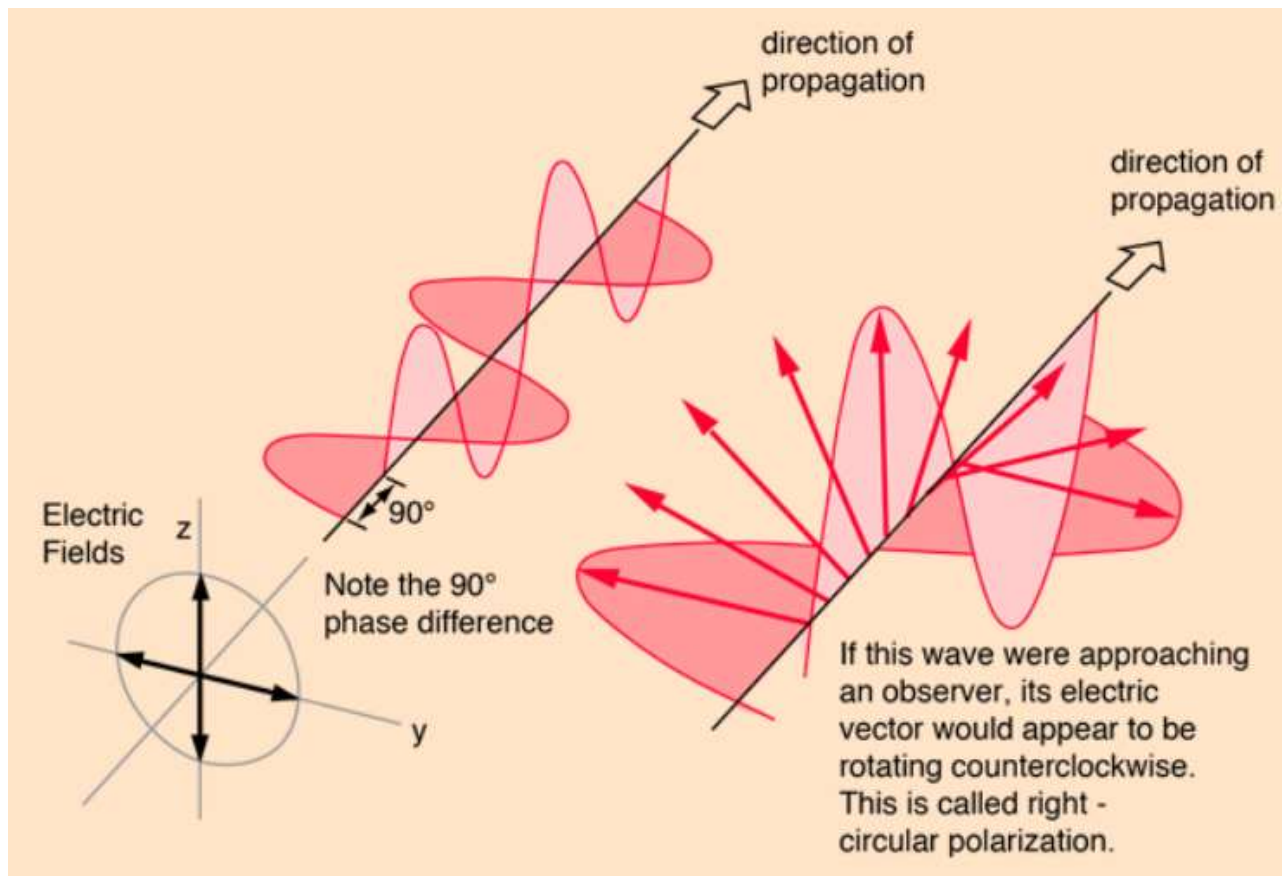
In linear polarization, the electric field of light is limited to a single plane along the direction of propagation.



Linear polarization

4.1.2 CIRCULAR POLARIZATION

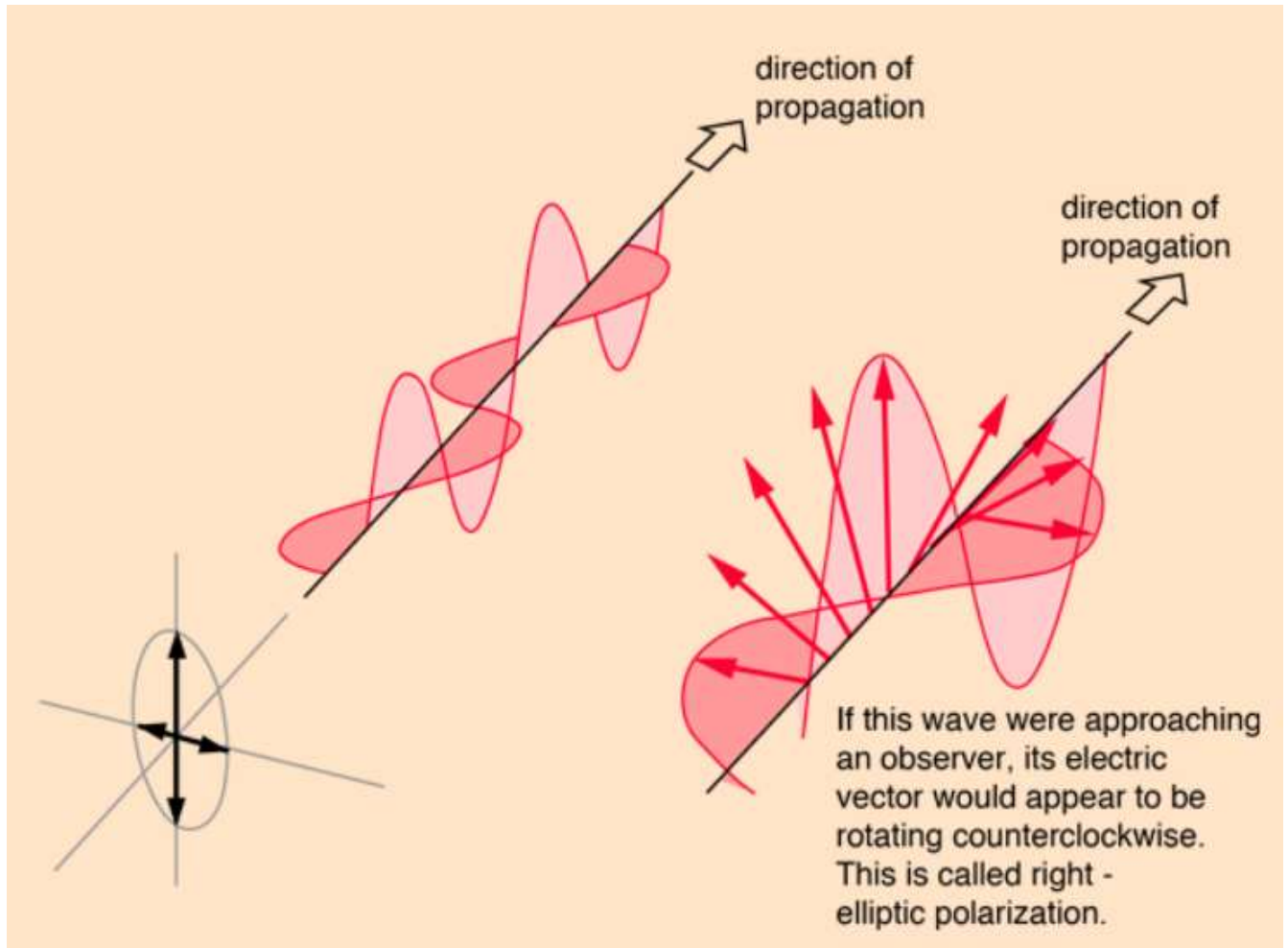
There are two linear components in the electric field of light that are perpendicular to each other such that their amplitudes are equal, but the phase difference is $\pi/2$. The propagation of the occurring electric field will be in a circular motion.



Circular Polarization

4.1.3 ELLIPTICAL POLARIZATION

The electric field of light follows an elliptical propagation. The amplitude and phase difference between the two linear components are not equal.



Elliptical Polarization

5 METHODS USED IN THE POLARIZATION OF LIGHT

There are a few methods used in the polarization of light:

- Polarization by Transmission
- Polarization by Reflection
- Polarization by Scattering
- Polarization by Refraction



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6 POLARIZATION APPLICATIONS

Following are the applications of polarization:

- Polarization is used in sunglasses to reduce the glare.
- Polaroid filters are used in plastic industries for performing stress analysis tests.
- Three-dimensional movies are produced and shown with the help of polarization.
- Polarization is used for differentiating between transverse and longitudinal waves.
- Infrared spectroscopy uses polarization.
- It is used in seismology to study earthquakes.
- In Chemistry, the chirality of organic compounds is tested using polarization techniques.